MEENAKSHI A

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Data Analytics and AI student with strong experience in Python, SQL, and large-scale data processing. Skilled in Natural Language Processing (NLP), machine learning, predictive modeling, and data visualization. Experienced in architecting hybrid deep learning pipelines, engineering semantic text embedding wrappers with Sentence-BERT, and deploying high-performance production models. Passionate about applying multi-modal analytical architectures to uncover complex behavioral and sustainability trends.

EDUCATION

Bachelor of Technology in Computer Science Engineering (Data Analytics) *09/2022 - 07/2026*

Alliance University, Bangalore **CGPA: 8.7 (Top 10% of class)**

Relevant Coursework: Data Structures, Machine Learning, Database Management, Business Intelligence, Data Mining, Data Visualization.

TECHNICAL SKILLS

Languages & Databases:	Python, SQL, C, C++, PostgreSQL, MySQL, MongoDB, Cassandra, NoSQL
Data Engineering & Cloud:	Apache Spark, Apache Kafka, Apache Hive, Hadoop, AWS, Google Cloud Platform (GCP), BigQuery, AI Platform
Machine Learning & AI:	CatBoost Classifier, Sentence-BERT (SBERT), Hybrid Architectures, Natural Language Processing (NLP), XGBoost, LightGBM, Scikit-learn, Pandas, NumPy, Hugging Face Spaces (Deployment)
Analytics & Visualization:	Power BI, Tableau, Excel, Google Data Studio, Matplotlib, Seaborn, Business Intelligence, Data Warehousing
Developer Tools:	Git, Jupyter Notebook, Data Structures & Algorithms

TECHNICAL PROJECTS

Analyzing Consumer Perception of Eco-Friendly & Sustainable Products *05/2026*

Alliance University, Bengaluru | Supervisor: Dr. Sridhar D | Final Year Capstone Project

- Developed an NLP-driven hybrid predictive framework combining deep learning feature engineering with a CatBoost classification model.
- Implemented Sentence-BERT (SBERT) to extract high-dimensional semantic text embeddings from unstructured consumer suggestion inputs.
- Engineered a multi-modal data pipeline in Python to merge semantic text vectors with vectorized demographic attributes and state-wise purchase frequencies.
- Optimized and deployed the final hybrid SBERT-CatBoost classifier as an interactive web service on Hugging Face Spaces to map consumer behavior trends in real time.

Improving Agricultural Resilience Using IoT Devices *Academic Project*

Alliance University

- Developed an IoT-based agricultural monitoring system using real-time sensor data to track soil moisture, temperature, and crop health.
- Built predictive analytics models in Python to forecast irrigation requirements, improving resource planning efficiency by 25% and enabling data-driven crop management decisions.

Real-time E-commerce Recommendation System *Personal Project*

- Built a real-time recommendation engine using Python, Apache Kafka, Spark Streaming, and HDFS to process large-scale customer activity datasets.
- Optimized distributed data pipelines to achieve sub-second recommendation delivery and deployed the end-to-end inference framework as a live application on Hugging Face Spaces, improving personalization transparency.

Climate TrendAI - Climate Data Analysis & Prediction Web App *Personal Project*

- Developed an interactive climate analytics web application using Python, Gradio, Pandas, and Scikit-learn to visualize historical temperature trends across countries and predict future climate patterns, achieving 92% forecasting accuracy through machine learning optimization; deployed live on Hugging Face Spaces.

Ransomware Simulator & Decryption Model *Personal Project*

- Simulated ransomware attack workflows using Python-based encryption techniques and developed an AI-powered decryption model for secure recovery, reducing simulated recovery time by 35% across multiple encryption scenarios.

CERTIFICATIONS

CertNexus Certified Data Science Practitioner | Machine Learning with Python - IBM | Business Intelligence (BI) Essentials - IBM | Algorithmic Toolbox - UC San Diego | Data Structures - UC San Diego